

Question 1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">
  <title>TechNova - Digital Experiences</title>
  <link rel="stylesheet" href="question 1.css">
</head>
<body>
  <!-- =====
  HEADER
  ===== -->

  <header class="header">
    <!-- Logo -->
    <div class="logo">
      <span class="logo-icon">◆</span>
      <span>TechNova</span>
    </div>

    <!-- Mobile Menu Checkbox -->
    <input type="checkbox"
      id="menu-toggle"
      class="menu-toggle">

    <!-- Hamburger Icon -->
    <label for="menu-toggle"
      class="menu-icon">

    </label>

    <!-- Navigation -->
    <nav class="navbar">
      <a href="#home">Home</a>
      <a href="#about">About</a>
      <a href="#services">Services</a>
      <a href="#portfolio">Portfolio</a>
      <a href="#contact">Contact</a>
    </nav>

    <!-- Get Started -->
    <a href="#contact"
      class="get-started">
      Get Started
    </a>
  </header>

  <!-- HOME / HERO-->
  <section class="hero">
```

```

        id="home">
<!-- Hero Text -->
<div class="hero-text">
    <h1>
        Build Better
    <span>
        Digital Experiences
    </span>
</h1>
<p>
    We help brands and businesses grow
    with modern, high-performance
    websites and applications.
</p>
<a href="#services"
    class="learn-button">
    Learn More
</a>
</div>
<!-- Hero Dashboard -->
<div class="hero-image">
    <div class="monitor">
        <!-- Monitor Header -->
        <div class="monitor-top">
            <span>
                Dashboard
            </span>
            <span>
                • • •
            </span>
        </div>
        <!-- Dashboard Content -->
        <div class="dashboard">
            <!-- Bar Chart -->
            <div class="chart">
                <div class="bar bar1"></div>
                <div class="bar bar2"></div>
                <div class="bar bar3"></div>
                <div class="bar bar4"></div>
                <div class="bar bar5"></div>
            </div>
            <!-- Circle -->
            <div class="circle-chart">
                <div class="circle-inner">

```

```

        </div>
    </div>
</div>
</div>
<!-- Plant -->
<div class="plant">
    🌿
</div>
</div>
</section>
<!-- =====
      ABOUT
      ===== -->

<section class="about"
      id="about">

    <h2>
        ABOUT US
    </h2>

    <p>

        TechNova is a creative team focused
        on building modern websites and
        digital experiences for businesses.

    </p>

</section>

<!-- =====
      SERVICES
      ===== -->

<section class="services"
      id="services">

    <h2>
        OUR SERVICES

```

```
</h2>
```

```
<div class="service-grid">
```

```
<!-- Service 1 -->
```

```
<div class="service-card">
```

```
<div class="service-icon blue">
```



```
</div>
```

```
<h3>
```

```
Web Development
```

```
</h3>
```

```
<p>
```

```
We build modern and  
responsive websites.
```

```
</p>
```

```
</div>
```

```
<!-- Service 2 -->
```

```
<div class="service-card">
```

```
<div class="service-icon green">
```



```
</div>
```

```
<h3>
```

```
    UI/UX Design
```

```
</h3>
```

```
<p>
```

```
    We create beautiful and  
    user-friendly designs.
```

```
</p>
```

```
</div>
```

```
<!-- Service 3 -->
```

```
<div class="service-card">
```

```
    <div class="service-icon purple">
```

```
        &lt;/&gt;
```

```
    </div>
```

```
<h3>
```

```
    Web Applications
```

```
</h3>
```

```
<p>
```

```
    We develop powerful and  
    scalable web applications.
```

```

        </p>

    </div>

</div>

</section>

<!-- =====
      OUR WORK / PORTFOLIO
      ===== -->

<section class="our-work"
      id="portfolio">

    <h2>
      OUR WORK
    </h2>

    <div class="work-grid">

        <!-- Image 1 -->

        <div class="work-item">

        </div>

        <!-- Image 2 -->

        <div class="work-item">

```

```
        
```

```
    </div>
```

```
<!-- Image 3 -->
```

```
<div class="work-item">
```

```
    
```

```
</div>
```

```
<!-- Image 4 -->
```

```
<div class="work-item">
```

```
    
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- =====
CONTACT / FOOTER
===== -->
```

```
<footer class="footer">
```

```
id="contact">
```

```
<div class="footer-grid">
```

```
<!-- Company -->
```

```
<div class="footer-column">
```

```
<h3>
```

```
TechNova
```

```
</h3>
```

```
<p>
```

```
We are a creative team  
building modern digital  
products.
```

```
</p>
```

```
<div class="social-icons">
```

```
<span>●</span>
```

```
<span>●</span>
```

```
<span>●</span>
```

```
<span>●</span>
```

```
</div>
```

```
</div>
```

```
<!-- Quick Links -->
```

```
<div class="footer-column">
```

```
<h3>
    Quick Links
</h3>
```

```
<a href="#home">
    Home
</a>
```

```
<a href="#about">
    About
</a>
```

```
<a href="#services">
    Services
</a>
```

```
<a href="#portfolio">
    Portfolio
</a>
```

```
<a href="#contact">
    Contact
</a>
```

```
</div>
```

```
<!-- Services -->
```

```
<div class="footer-column">
```

```
<h3>
    Services
</h3>
```

```
<a href="#services">
    Web Development
```

```
</a>
```

```
<a href="#services">
```

```
  UI/UX Design
```

```
</a>
```

```
<a href="#services">
```

```
  Web Applications
```

```
</a>
```

```
<a href="#services">
```

```
  SEO Optimization
```

```
</a>
```

```
</div>
```

```
<!-- Newsletter -->
```

```
<div class="footer-column newsletter">
```

```
<h3>
```

```
  Newsletter
```

```
</h3>
```

```
<p>
```

```
  Subscribe to get  
  updates and news.
```

```
</p>
```

```
<input
```

```
  type="email"
```

```
  placeholder="Your email">
```

```
<button>
```

```
        Subscribe
    </button>

</div>

</div>

<!-- Copyright -->

<div class="copyright">

    © 2026 TechNova.
    All rights reserved.

</div>

</footer>

</body>

</html>
```

Question1.css

```
/* =====
GENERAL
===== */

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}
```

```
body {  
    font-family: Arial, sans-serif;  
  
    background-color: white;  
  
    color: #172033;  
  
    line-height: 1.6;  
}  
  
a {  
    text-decoration: none;  
}  
  
img {  
    max-width: 100%;  
}  
  
/* =====  
    HEADER  
    ===== */  
  
.header {  
  
    width: 100%;  
  
    min-height: 65px;  
  
    background-color: #0b1d32;  
  
    color: white;  
  
    display: grid;  
  
    grid-template-columns: 1fr auto auto;  
  
    align-items: center;  
  
    padding: 0 35px;  
  
    position: sticky;
```

```
    top: 0;

    z-index: 1000;
}

/* Logo */

.logo {

    display: flex;

    align-items: center;

    gap: 7px;

    font-size: 18px;

    font-weight: bold;
}

.logo-icon {

    font-size: 14px;

    color: white;
}

/* =====
NAVIGATION
===== */

.navbar {

    display: flex;

    align-items: center;

    gap: 25px;

    margin-right: 30px;
}
```

```
.navbar a {

    color: white;

    font-size: 12px;

    transition: 0.3s;
}

.navbar a:hover {

    color: #4285e8;
}

/* =====
GET STARTED
===== */

.get-started {

    background-color: #1769e0;

    color: white;

    padding: 9px 16px;

    border-radius: 3px;

    font-size: 11px;

    transition: 0.3s;
}

.get-started:hover {

    background-color: #0e56bd;
}

/* =====
MOBILE MENU
===== */
```

```
.menu-toggle {  
  
    display: none;  
}  
  
.menu-icon {  
  
    display: none;  
  
    font-size: 25px;  
  
    cursor: pointer;  
}  
  
/* =====  
HERO  
===== */  
  
.hero {  
  
    min-height: 310px;  
  
    padding: 35px 45px;  
  
    display: grid;  
  
    grid-template-columns: 1fr 1fr;  
  
    align-items: center;  
  
    gap: 30px;  
  
    background-color: white;  
}  
  
/* Hero Text */  
  
.hero-text {  
  
    padding-left: 15px;  
}
```

```
.hero-text h1 {

    font-size: 28px;

    line-height: 1.2;

    margin-bottom: 15px;
}

.hero-text h1 span {

    display: block;

    color: #1769e0;
}

.hero-text p {

    width: 80%;

    color: #596579;

    font-size: 13px;

    line-height: 1.6;

    margin-bottom: 18px;
}

/* Learn More */

.learn-button {

    display: inline-block;

    background-color: #1769e0;

    color: white;

    padding: 10px 20px;

    border-radius: 3px;
```

```

        font-size: 12px;

        transition: 0.3s;
    }

.learn-button:hover {

    background-color: #0e56bd;
}

/* =====
DASHBOARD
===== */

.hero-image {

    min-height: 230px;

    display: flex;

    align-items: center;

    justify-content: center;

    position: relative;
}

/* Monitor */

.monitor {

    width: 285px;

    height: 165px;

    border: 7px solid #25364c;

    border-radius: 6px;

    background-color: #f4f8fc;

```

```
    position: relative;

    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.15);
}
```

```
/* Monitor top */
```

```
.monitor-top {

    height: 23px;

    background-color: #dce8f5;

    padding: 5px 8px;

    font-size: 7px;

    display: flex;

    justify-content: space-between;
}
```

```
/* Dashboard */
```

```
.dashboard {

    height: 128px;

    display: grid;

    grid-template-columns: 1fr 1fr;

    align-items: center;

    padding: 15px;
}
```

```
/* Chart */
```

```
.chart {

    height: 90px;
}
```

```
    display: flex;

    align-items: flex-end;

    gap: 7px;
}

.bar {

    width: 17px;

    background-color: #4285e8;

    border-radius: 2px 2px 0 0;
}

.bar1 {
    height: 35px;
}

.bar2 {
    height: 55px;
}

.bar3 {
    height: 45px;
}

.bar4 {
    height: 75px;
}

.bar5 {
    height: 62px;
}

/* Circle */
```

```
.circle-chart {  
  
    width: 75px;  
  
    height: 75px;  
  
    border-radius: 50%;  
  
    border: 11px solid #5a9bf5;  
  
    display: flex;  
  
    align-items: center;  
  
    justify-content: center;  
  
    margin: auto;  
}
```

```
.circle-inner {  
  
    font-size: 12px;  
  
    font-weight: bold;  
  
    color: #172033;  
}
```

```
/* Plant */
```

```
.plant {  
  
    position: absolute;  
  
    bottom: 20px;  
  
    left: 50px;  
  
    font-size: 55px;  
}
```

```
/* =====
```

```
ABOUT
===== */

.about {

    padding: 35px 45px;

    text-align: center;

    background-color: #f8fafc;
}

.about h2 {

    font-size: 10px;

    margin-bottom: 12px;

    color: #25364c;
}

.about p {

    max-width: 650px;

    margin: auto;

    color: #667085;

    font-size: 12px;
}

/* =====
SERVICES
===== */

.services {

    padding: 25px 45px 30px;
}

.services h2,
```

```
.our-work h2 {

    text-align: center;

    font-size: 10px;

    margin-bottom: 18px;

    color: #25364c;
}

/* Service Grid */

.service-grid {

    display: grid;

    grid-template-columns: repeat(3, 1fr);

    gap: 20px;
}

/* Service Card */

.service-card {

    border: 1px solid #e2e6eb;

    border-radius: 5px;

    padding: 20px;

    text-align: center;

    min-height: 120px;

    box-shadow: 0 2px 5px rgba(0, 0, 0, 0.04);

    transition: 0.3s;
}

.service-card:hover {
```

```
    transform: translateY(-3px);

    box-shadow: 0 5px 12px rgba(0, 0, 0, 0.08);
}
```

```
.service-icon {

    font-size: 22px;

    margin-bottom: 7px;
}
```

```
.blue {

    color: #1769e0;
}
```

```
.green {

    color: #42a889;
}
```

```
.purple {

    color: #8c6de8;
}
```

```
.service-card h3 {

    font-size: 11px;

    margin-bottom: 7px;
}
```

```
.service-card p {

    color: #667085;

    font-size: 9px;
}
```

```
}

/* =====
OUR WORK
===== */

.our-work {

    padding: 10px 45px 35px;
}

/* Work Grid */

.work-grid {

    display: grid;

    grid-template-columns: repeat(4, 1fr);

    gap: 10px;
}

/* Image Container */

.work-item {

    height: 130px;

    overflow: hidden;

    border-radius: 4px;

    background-color: #eeeeee;
}

/* Image */

.work-item img {

    width: 100%;

    height: 100%;
}
```

```
    object-fit: cover;

    display: block;

    transition: 0.3s;
}

.work-item:hover img {

    transform: scale(1.05);
}

/* =====
FOOTER
===== */

.footer {

    background-color: #0b1d32;

    color: white;

    padding: 25px 45px 12px;
}

/* Footer Grid */

.footer-grid {

    display: grid;

    grid-template-columns: 1.4fr 1fr 1fr 1.4fr;

    gap: 35px;
}

/* Footer Column */

.footer-column {

    display: flex;
```

```
    flex-direction: column;

    gap: 6px;
}
```

```
.footer-column h3 {

    font-size: 11px;

    margin-bottom: 6px;
}
```

```
.footer-column p,
.footer-column a {

    color: #c5ced9;

    font-size: 8px;

    line-height: 1.5;
}
```

```
.footer-column a:hover {

    color: white;
}
```

```
/* Social */
```

```
.social-icons {

    display: flex;

    gap: 8px;

    margin-top: 8px;
}
```

```
.social-icons span {
```

```
    font-size: 10px;
}

/* Newsletter */

.newsletter input {

    border: none;

    outline: none;

    padding: 8px;

    width: 100%;

    font-size: 8px;

    margin-top: 3px;
}

.newsletter button {

    border: none;

    background-color: #1769e0;

    color: white;

    padding: 8px;

    font-size: 8px;

    cursor: pointer;
}

.newsletter button:hover {

    background-color: #0e56bd;
}

/* Copyright */
```

```
.copyright {

    text-align: center;

    border-top: 1px solid #2c3b4c;

    margin-top: 18px;

    padding-top: 10px;

    font-size: 8px;

    color: #9da9b8;
}

/* =====
TABLET
===== */

@media screen and (max-width: 900px) {

    /* Header */

    .header {

        grid-template-columns: 1fr auto;

        padding: 0 25px;
    }

    .navbar {

        display: none;

        position: absolute;

        top: 65px;

        left: 0;

        width: 100%;
    }
}
```

```
background-color: #0b1d32;

flex-direction: column;

gap: 0;

margin: 0;

padding: 10px 0;
}

.navbar a {

width: 100%;

text-align: center;

padding: 12px;
}

.menu-icon {

display: block;

grid-column: 2;

grid-row: 1;
}

.get-started {

display: none;
}

/* Mobile menu opens */

.menu-toggle:checked ~ .navbar {

display: flex;
}
```

```
/* Hero */

.hero {

    grid-template-columns: 1fr 1fr;

    padding: 30px;
}

.hero-text h1 {

    font-size: 24px;
}

/* Services */

.service-grid {

    grid-template-columns: repeat(2, 1fr);
}

/* Work */

.work-grid {

    grid-template-columns: repeat(2, 1fr);
}

.work-item {

    height: 150px;
}

/* Footer */

.footer-grid {

    grid-template-columns: repeat(2, 1fr);
}
}
```

```
/* =====  
MOBILE  
===== */
```

```
@media screen and (max-width: 600px) {
```

```
/* Header */
```

```
.header {
```

```
    min-height: 55px;
```

```
    padding: 0 15px;
```

```
}
```

```
.logo {
```

```
    font-size: 14px;
```

```
}
```

```
.navbar {
```

```
    top: 55px;
```

```
}
```

```
/* Hero */
```

```
.hero {
```

```
    grid-template-columns: 1fr;
```

```
    padding: 30px 20px;
```

```
    text-align: left;
```

```
}
```

```
.hero-text {
```

```
    padding-left: 0;
```

```
}

.hero-text h1 {

    font-size: 24px;
}

.hero-text p {

    width: 100%;

    font-size: 11px;
}

.hero-image {

    min-height: 210px;

    margin-top: 10px;
}

.monitor {

    width: 250px;

    height: 145px;
}

.plant {

    left: 20px;

    bottom: 15px;

    font-size: 45px;
}

/* About */

.about {
```

```
padding: 30px 20px;
}

/* Services */

.services {

padding: 25px 20px 30px;
}

.service-grid {

grid-template-columns: 1fr;

gap: 12px;
}

.service-card {

min-height: 110px;
}

/* Portfolio */

.our-work {

padding: 10px 20px 30px;
}

.work-grid {

grid-template-columns: 1fr 1fr;

gap: 8px;
}

.work-item {

height: 125px;
```

```

}

/* Footer */

.footer {

    padding: 25px 20px 12px;
}

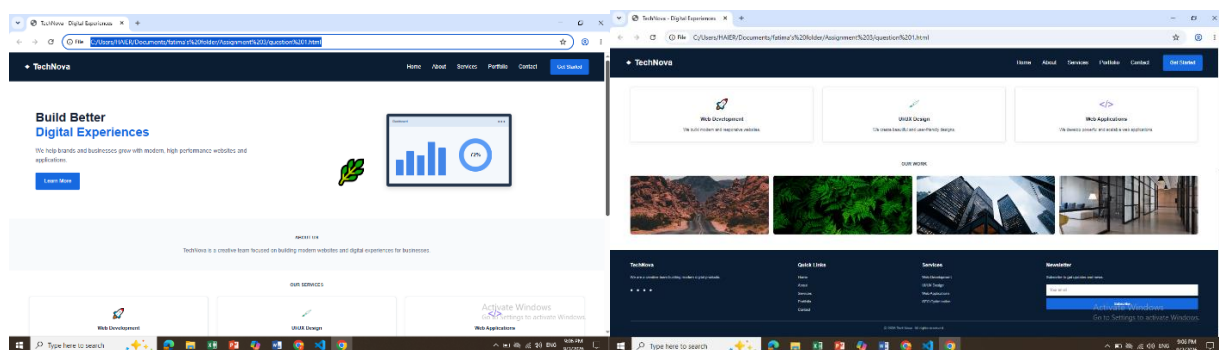
.footer-grid {

    grid-template-columns: 1fr 1fr;

    gap: 25px;
}
}

```

Question1 Output:



Question2.html

```

<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport"
        content="width=device-width, initial-scale=1.0">

    <title>DOM Manipulation</title>

```

```

</head>

<body>

  <h1 id="title">
    Welcome to JavaScript DOM
  </h1>

  <p id="message">
    JavaScript can access and change HTML elements using the DOM.
  </p>

  <button id="showBtn">
    Show Information
  </button>

  <script src="question2.js"></script>

</body>

</html>

```

Question2.js

```

// Access the heading using getElementById()
var title = document.getElementById("title");

// Access the paragraph using getElementById()
var message = document.getElementById("message");

// Access the button using getElementById()
var showBtn = document.getElementById("showBtn");

// Use document.write() to display a simple message
document.write("<p>This is a message displayed using document.write().</p>");

// When the button is clicked
showBtn.onclick = function() {

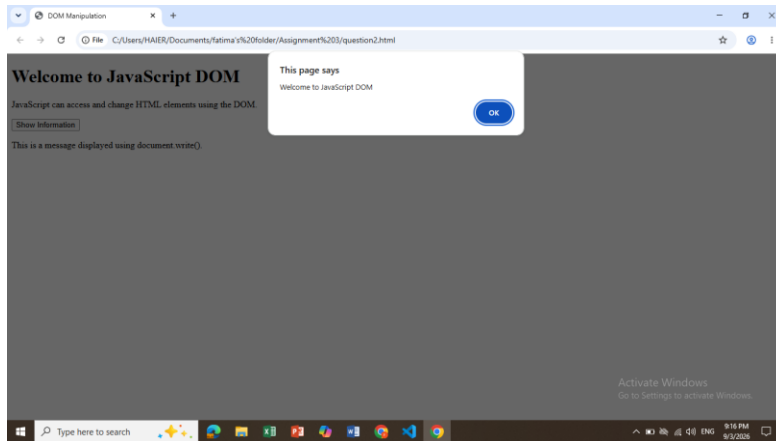
  // Display the heading text using alert()
  alert(title.innerText);
}

```

```
// Display the paragraph text using console.log()
console.log(message.innerText);

};
```

Question2 Output:



Question 3.html

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">

  <title>JavaScript Variables and Data Types</title>

</head>

<body>

  <h1>JavaScript Variables and Data Types</h1>

  <p>
    Open the browser console to see the variables,
    their values, and their data types.
  </p>
```

```
<script src="q3.js"></script>

</body>

</html>
```

Question3.js

```
// =====
// VARIABLES USING var, let AND const
// =====

// var
// var is used to declare a variable.
var studentName = "Fatima";

// let
// let is used to declare a variable
// whose value can be changed.
let age = 20;

// const
// const is used to declare a variable
// whose value cannot be reassigned.
const university = "University of Sindh";

console.log("var studentName:", studentName);
console.log("let age:", age);
console.log("const university:", university);

// =====
// 1. STRING
// =====

// String represents text.

var name = "Ali";

console.log("String value:", name);
console.log("String type:", typeof name);
```

```
// =====  
// 2. NUMBER  
// =====  
  
// Number represents numeric values.  
  
var marks = 85;  
  
console.log("Number value:", marks);  
console.log("Number type:", typeof marks);  
  
// =====  
// 3. BOOLEAN  
// =====  
  
// Boolean represents true or false.  
  
var passed = true;  
  
console.log("Boolean value:", passed);  
console.log("Boolean type:", typeof passed);  
  
// =====  
// 4. UNDEFINED  
// =====  
  
// Undefined means a variable has been declared  
// but no value has been assigned to it.  
  
var result;  
  
console.log("Undefined value:", result);  
console.log("Undefined type:", typeof result);  
  
// =====  
// 5. NULL  
// =====  
  
// Null represents an intentional empty value.  
  
var data = null;
```

```
console.log("Null value:", data);
console.log("Null type:", typeof data);

// =====
// 6. OBJECT
// =====

// Object stores information using
// key-value pairs.

var student = {
  name: "Sara",
  age: 21,
  course: "IT"
};

console.log("Object value:", student);
console.log("Object type:", typeof student);

// =====
// 7. ARRAY
// =====

// Array stores multiple values
// in a single variable.

var subjects = [
  "HTML",
  "CSS",
  "JavaScript"
];

console.log("Array value:", subjects);
console.log("Array type:", typeof subjects);

// =====
// 8. SYMBOL
// =====

// Symbol creates a unique value.

var studentID = Symbol("studentID");
```

```

console.log("Symbol value:", studentID);
console.log("Symbol type:", typeof studentID);

// =====
// 9. BIGINT
// =====

// BigInt is used to represent
// very large integer values.

var largeNumber = 12345678901234567890n;

console.log("BigInt value:", largeNumber);
console.log("BigInt type:", typeof largeNumber);

```

Question 3 Output:

